Lab 1.2

Time of Flight in Coax Cables

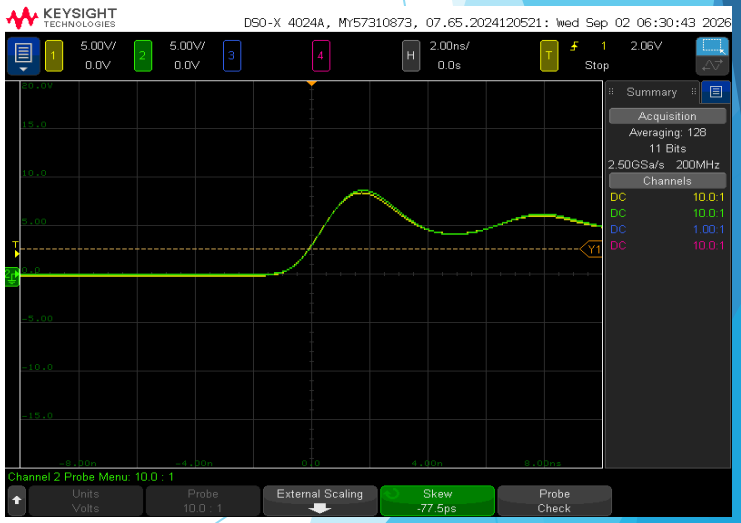
Leo Marcinkus and Jordan Bernhardt

Setup

- Scope
  - Keysight DSOX4024A
  - Dual channel acquisition, two 10x probes compensated on the calibration signal
  - Probes plug into BNC Tees through probe-to-coax adaptors
- Coax cable and adapters
  - RG58, Z_0 = 50 Ω
  - 4 × 6 ft lengths, SMA to BNC, BNC Tee adapters, with the far end open
- Test board
  - 555 timer driving a hex inverter at 5 V, using the 3 Ω output impedance
- **Equivalent circuit used to read every shot**
  - Source: V_{th} ≈ 5 V, R_{th} ≈ 3 Ω (1.71 Ω), rise ≈ 1.3 ns (limited by scope)
  - Cable: 50 Ω transmission line, TD = (delay per ft) x L

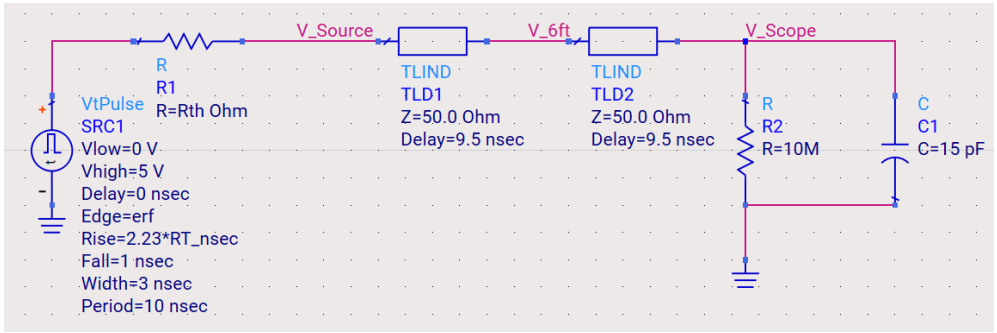


Board under test

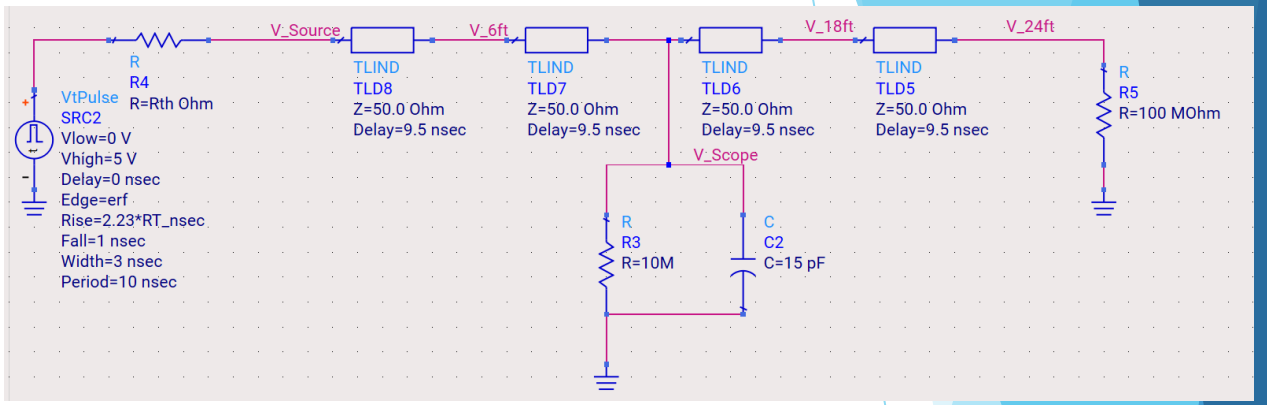


10x probe compensation

Measurement Method



Model of measurement at 12 ft terminated at scope

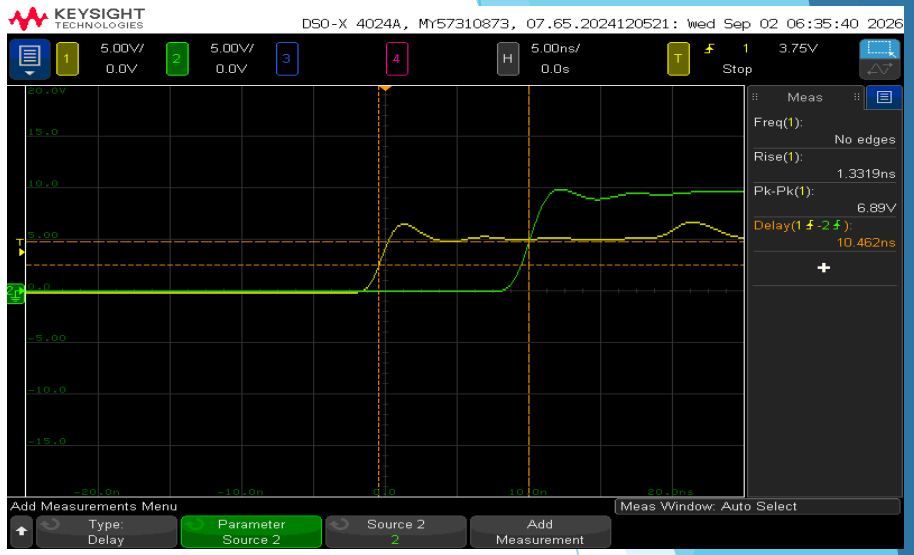


Model of measurement at 12 ft with all cables connected

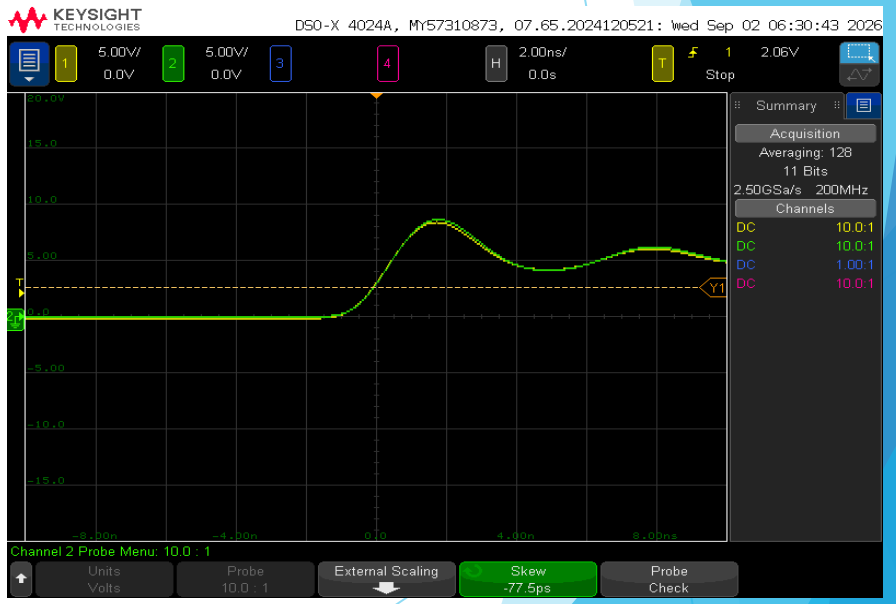
- Ch 1 at the launch Tee is the trigger and reference while Ch 2 moves. Delay = scope Delay (from Ch 1 to Ch 2), from 50 % to 50 %, with 128 averages
- **Incremental:** Ch 2 at the open end as a 6 ft cable is added each time (up to 24 ft)
- **All 24 ft connected:** Ch 2 moved Tee to Tee at 6, 12, 18 ft
- Two methods so Delay vs. Length is checked independently of how the cable is assembled
- **Skew Adjustment:** Both probes on the same edge, Ch 2 skew adjusted to -77.5 ps until the edges overlap

Measurement Uncertainties

- Physical
  - Cable lengths - +/- 2% (>2 in. per 6ft)
  - Connectors – add about 1 inch per cable
  - Trace lengths – << 1 in. total
  - Cable specs – manufacturing tolerance
- Acquisition
  - Scope resolution: vertical (~156mV), horizontal (~400 ps), and bandwidth (rise time)
    - Probe compensation by eye
  - Connectors and adapters – additional RLC from each



6 ft delay measurement: 10.462 ns



Skew adjustment: both probes on the launch edge

Data Collection

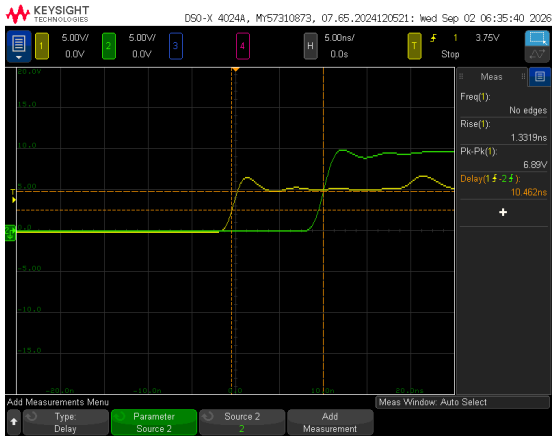
Propagation Delay vs. Cable Length (RG58) - Lab 1.2

Length (ft)	Incremental (ns)	Full 24 ft (ns)
6	10.462	10.303
12	19.915	19.864
18	29.361	29.324
24	38.888	

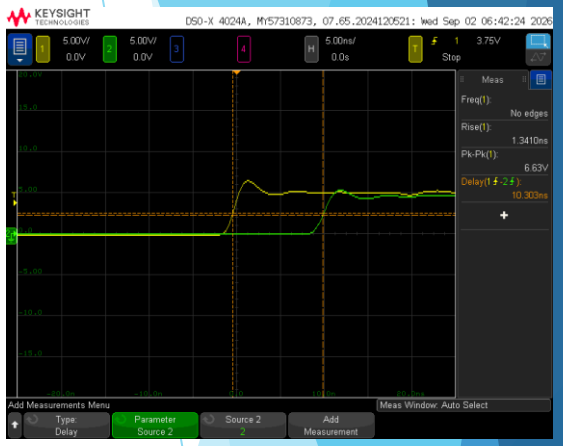
Linear Fit Results

Method	Slope (ns/ft)	Intercept (ns)	R^2
Incremental	1.5787	0.976	0.9999960854
Full 24 ft	1.5851	0.809	0.9999906017

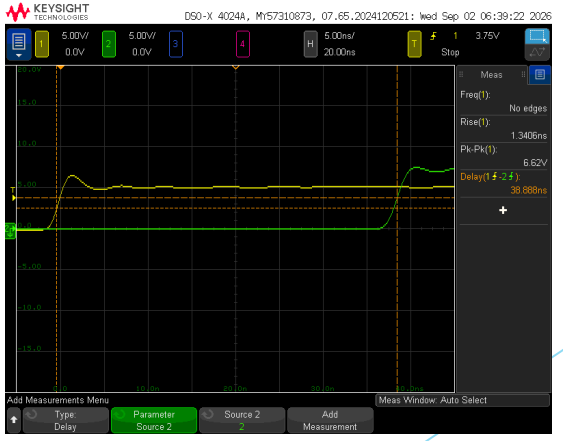
- Delay measurement on scope was used for values in data collection
  - Keysight uses 50% of amplitude and the time difference between the peaks
- Excel was used to format data and fit the lines
- Ch 1 rise time reads 1.33 ns in every capture. The source does not know how long the cable is



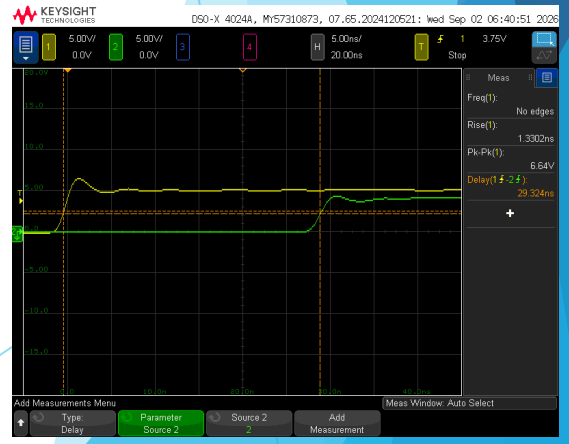
6 ft incremental measurement: 10.462 ns



6 ft with full 24 ft connected: 10.303 ns

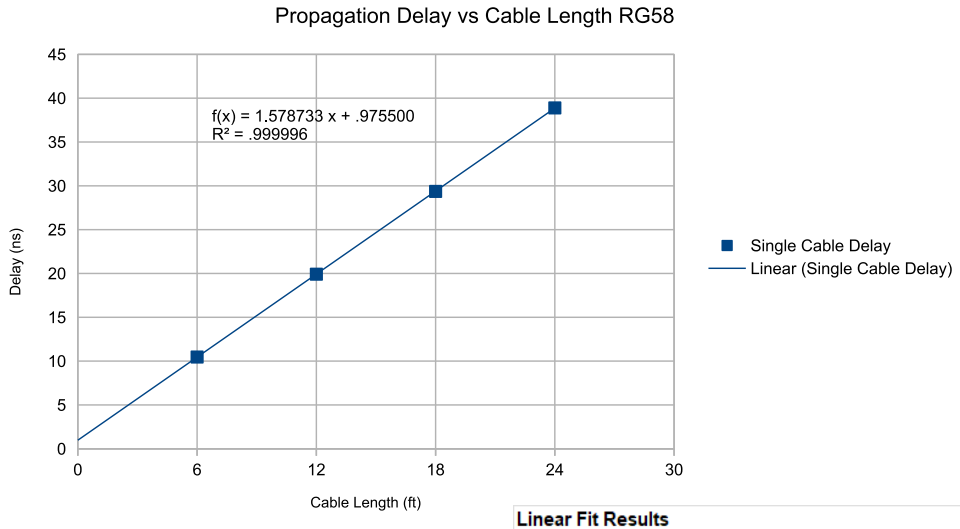


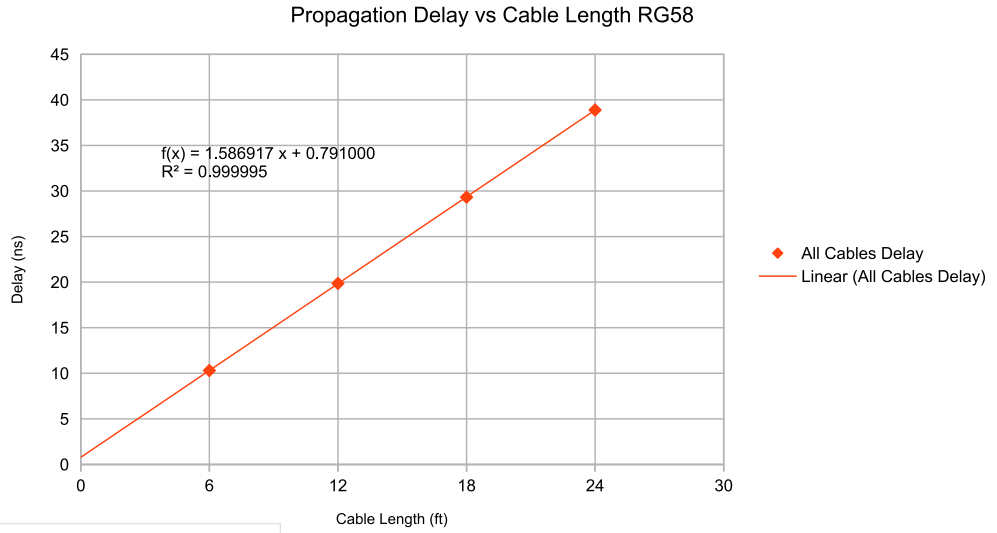
24 ft incremental: 38.888 ns



18 ft with full 24 ft connected: 29.324 ns

Data Analysis





Linear Fit Results

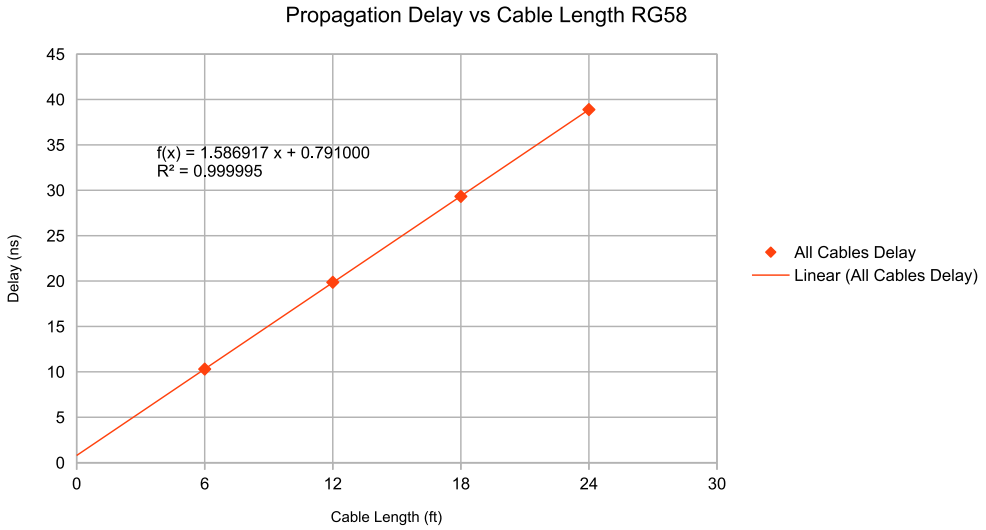
Method	Slope (ns/ft)	Intercept (ns)	R^2
Incremental	1.5787	0.976	0.9999960854
Full 24 ft	1.5851	0.809	0.9999906017

- **Model:** Delay = (ns/ft) · L + t_0
  - Linear relationship since every additional foot adds the same delay
- The slopes differ by less than 1 % and the intercepts differ by 0.17 ns. Both methods measure the same cable
- R^2 – measure of how well the line fits
- Absolute Percentage Error (A.P.E) and Mean Absolute Percentage error (M.A.P.E.) – ballpark of error for interpolated points

Length (Ft)	Single Cable Delay	A.P.E, Single cable	All Cables Delay	A.P.E
6	10.462	0.0013	10.303	0.0009
12	19.915	0.0003	19.864	0.0015
18	29.361	0.0011	29.324	0.0011
24	38.888	0.0006	38.888	0.0003
	M.A.P.E., Single Cable	0.0821	M.A.P.E., All Cables	0.0947

What the Intercept Means

- The intercept is the delay in the setup that is not related to the cable's length
  - Signal passes from the SMA-to-BNC adapter, through the BNC Tee connectors and probe adaptors, all before it reaches the cable
- It is wrong to assume there is 0 delay at a length of 0. Here $t_0 \approx 0.8$–1.0 ns, about the same as 7 inches of cable
- **Slope vs intercept:** the slope (1/velocity) is a property of the cable. The intercept is a property of everything in-between and differs by 0.17 ns between methods.
- With only two lengths you could not separate slope from intercept. Four lengths pin both down



Linear Fit Results			
Method	**Slope (ns/ft)**	**Intercept (ns)**	**R²**
Incremental	1.5787	0.976	0.9999960854
Full 24 ft	1.5851	0.809	0.9999906017

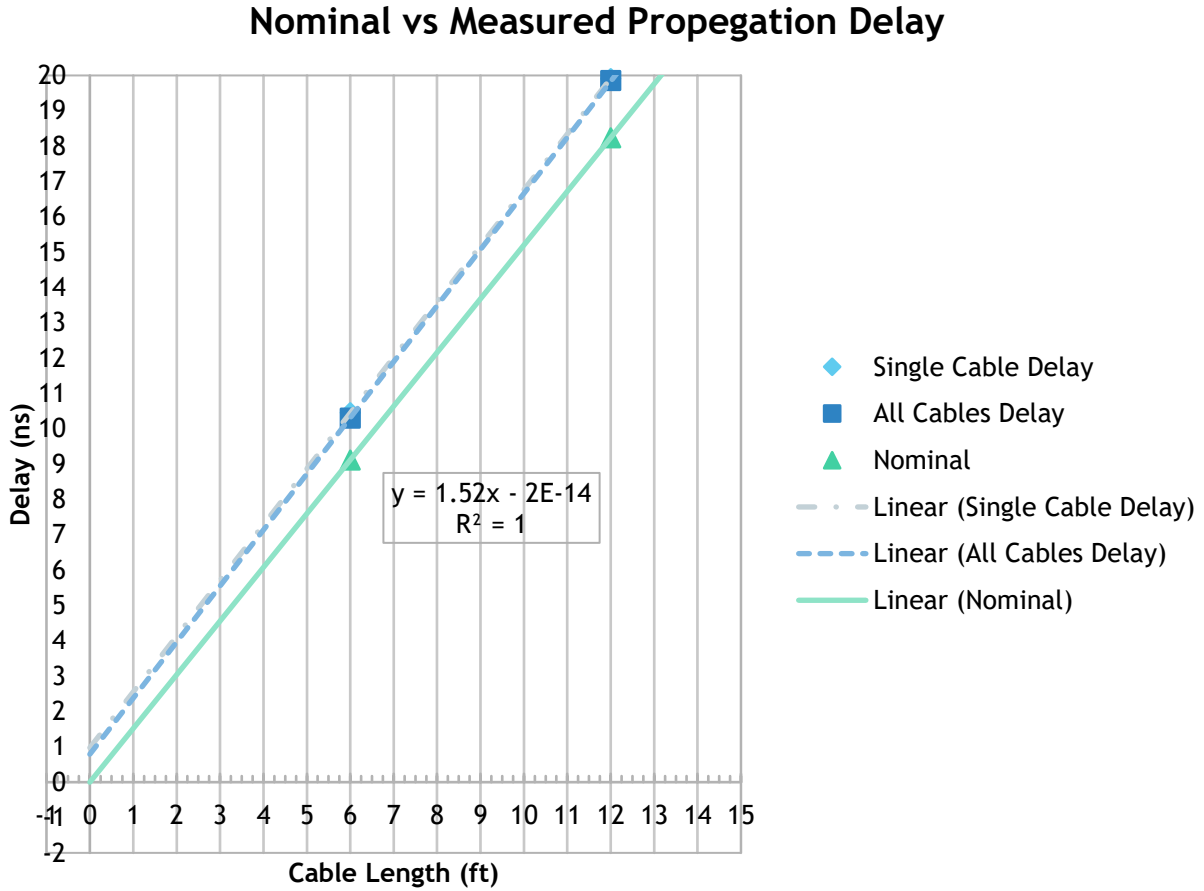
Expectations vs Reality

$$v = \frac{c}{\sqrt{\varepsilon_r}}$$

$v = c / \sqrt{\varepsilon_r}$ -> 1.54 ns/ft for $\varepsilon_r = 2.3$

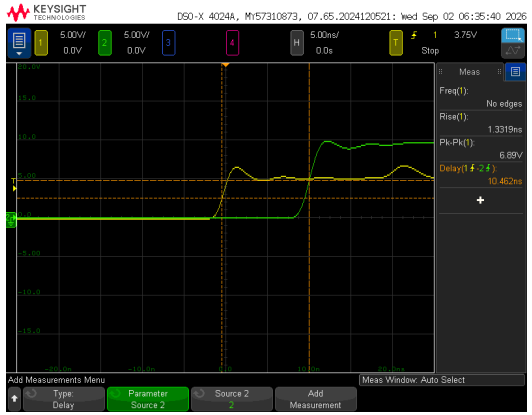
- **RG58 datasheet**
  - Capacitance 100 pF/m (± 5), i.e. 30.48 pF/ft
  - Characteristic impedance 50 Ω (± 3)
  - ε_r = 2.3 for polyethylene, velocity factor 0.66
- **Expected propagation speed:** $v = c/\sqrt{\varepsilon_r}$ = 0.66 c -> 5.06 ns/m or 1.54 ns/ft (0.649 ft/ns).
  - Cross-check: $Z_0 \cdot C$ = 50 Ω × 30.48 pF = 1.52 ns/ft
- **Turning the measurement into a Dk:** $\varepsilon_r = (c \cdot \text{slope})^2$ = (0.9836 ft/ns × 1.5822 ns/ft)² = 2.42, using the slope fitted through all seven points
- Measured 2.7 % slower than spec. The ± 5 % capacitance tolerance alone could allow a ± 2.5 % in delay., Each tee also adds a little length-proportional delay, so measurement and spec are consistent

Measured Cable Characteristics vs Specifications		
Speed of light (ft/ns):	0.9836	
Measured velocity (ft/ns):	0.6320	1/slope
Velocity Factor (measured):	0.6426	velocity of cable/C (in ft/ns)
Dk (measured):	2.422	1/VF^2
RG58 Spec (reference)		
Velocity Factor (spec):	0.66	
Spec delay per foot (ns/ft):	1.5404	
% difference (measured vs spec):	2.7%	

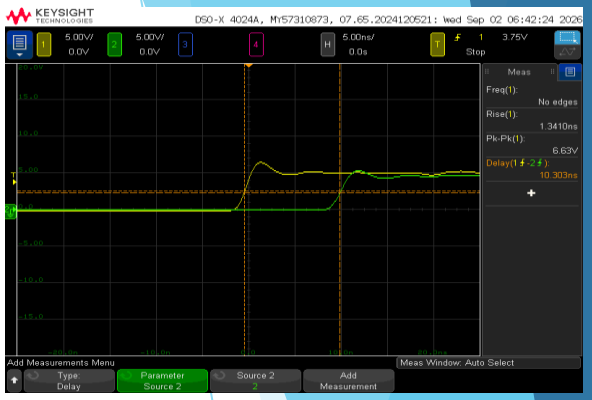


So What?

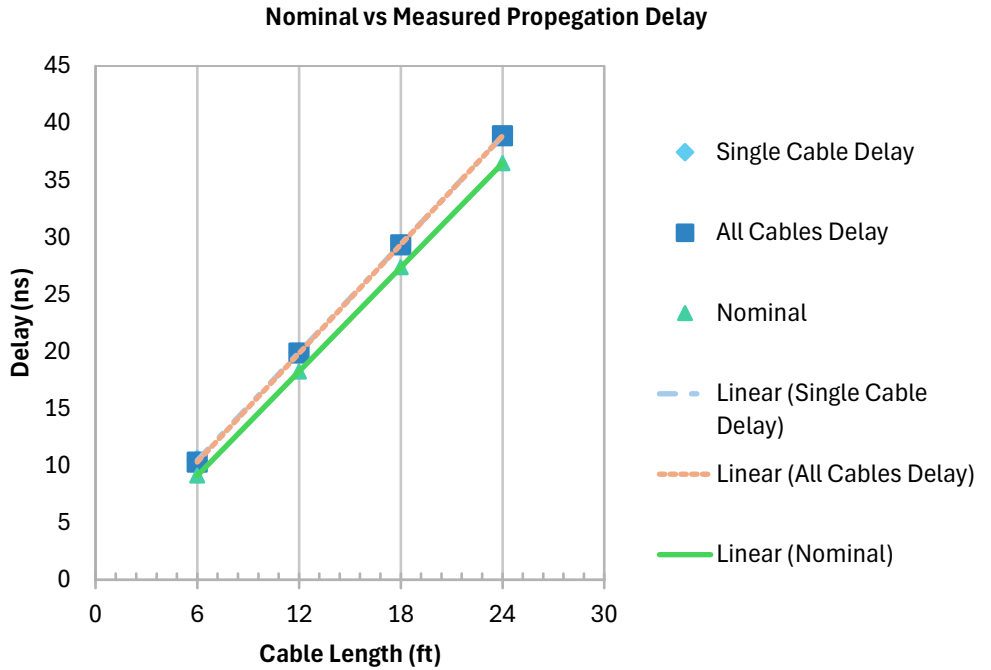
- **Signals take time to propagate, depending on the material:** 1.58 ns per foot of RG58, 9.5 ns per 6 ft section
- **Two different measurements can lead to the similar results:** incremental and all connected lead to the same results within uncertainty and error
- **Any measurement has uncertainty (and error):** the two methods disagree on the fixture delay by 0.17 ns. In an ideal environment, they would be exactly the same.
- **Consistency tests:** Ch 2 reads ≈ 4.8 V at a mid-cable Tee, the $V_{launched}$ from Lab 1.1, and ≈ 7 V at the open end where the reflection adds. On the 6 ft shot Ch 1 shows that reflection returning at ≈ 21 ns = 2 × 10.46 ns
- **Real world applicability:** Trace length matching in differential signaling. A 100 ps skew is only 0.06 ft (0.76 in) of this cable, and every connector counts at ≈0.2-0.4 ns



6 ft: Ch2 ≈ 7 V at the open end; Ch1 bump at ≈ 21 ns is the reflection returning

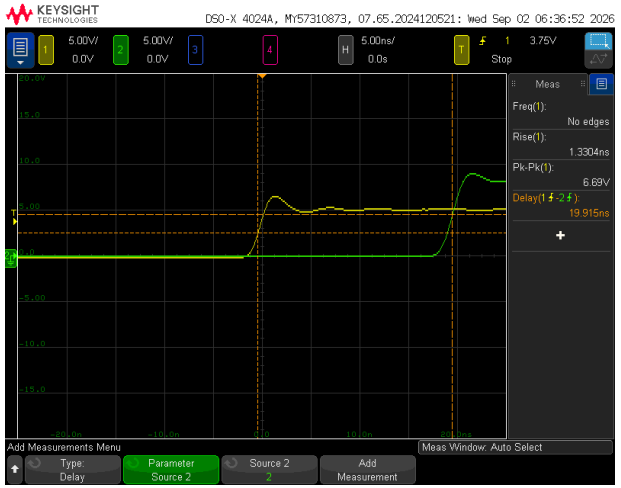


6 ft Tee mid-cable: Ch2 ≈ 4.8 V, no doubling, no bump

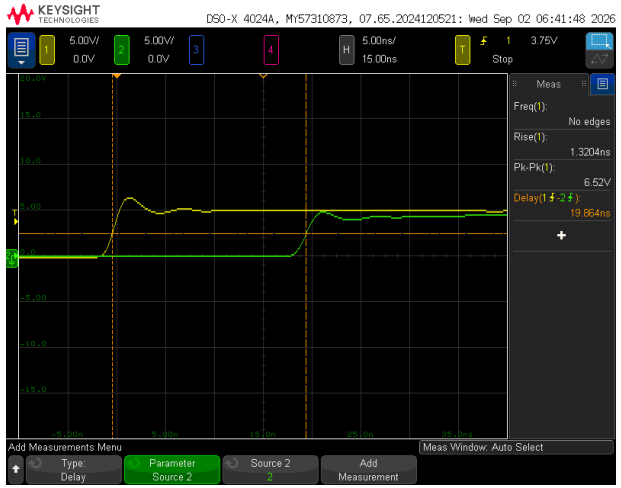


Questions?

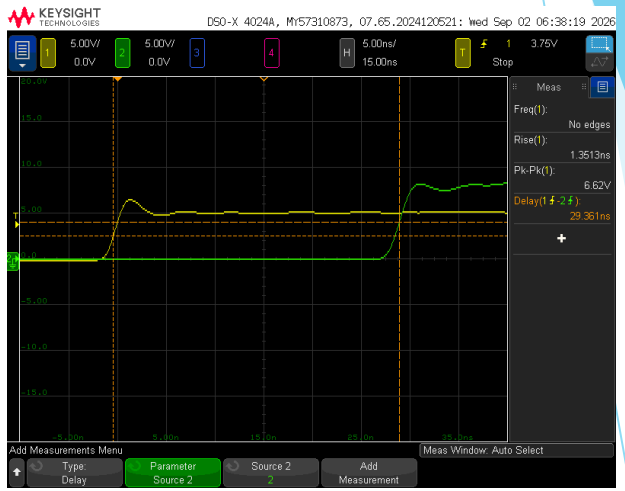
Additional data collections



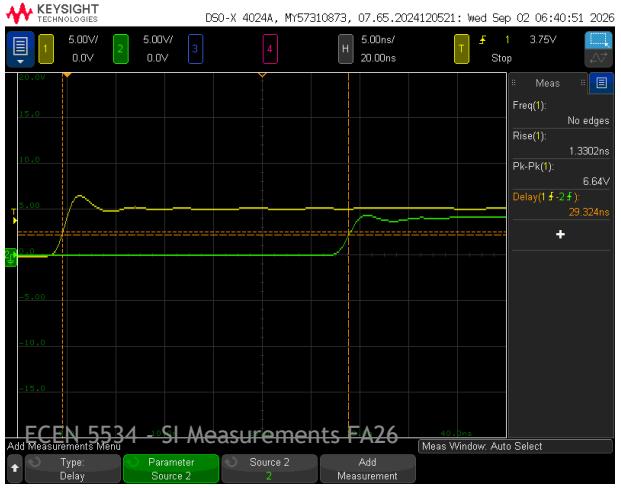
12 ft single



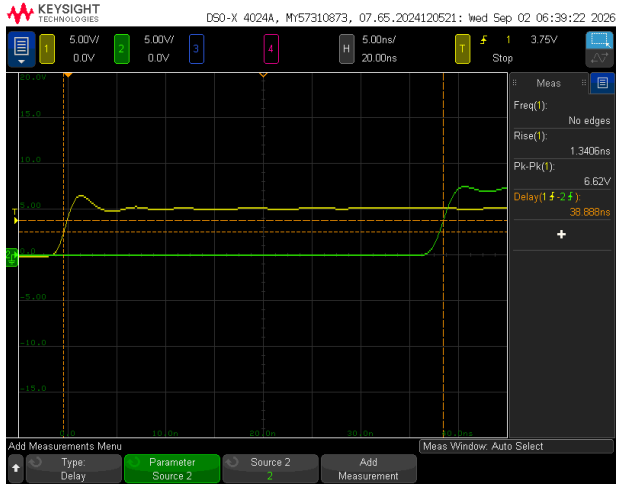
12 ft all



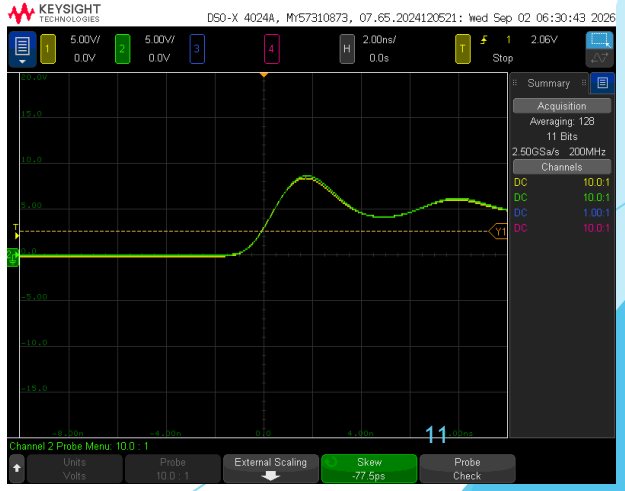
18 ft single



18 ft all



24 ft



Skew compensation

Additional data

Length (Ft)	Single Cable Delay	Absolute Error Single cable	All Cables Delay	Abs error All	Nominal
6	10.462	0.014102	10.303	0.144898	9.12
12	19.915	0.005296	19.864	0.056296	18.24
18	29.361	0.031694	29.324	0.068694	27.36
24	38.888	0.022908	38.888	0.022908	36.48
	Mean Error Single Cable	0.0185	Mean Error All Cable	0.073199	